



INSTITUTE OF SPACE TECHNOLOGY
KICSIT, Kahuta Campus



Assignment No 01

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Program:

BSCS-VI-B

Subject:

CS(Lab)

Submitted TO:

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Title: Network Security Tools & Analysis

Question No 1: Network Reconnaissance using Nmap

Objective

The goal of this task is to identify active services on a newly connected server using IP address **192.168.10.6**. This includes identifying open ports, running services, and operating system details.

Tool Used Nmap

Procedure

Step 1: Basic Scan

```
nmap 192.168.10.6
```

```
(kali@kali)-[~]
$ nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.5 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select

(kali@kali)-[~]
$ nmap 192.168.10.6
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-06 09:25 EDT
Nmap scan report for 192.168.10.6
Host is up (0.0022s latency).
All 1000 scanned ports on 192.168.10.6 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.60 seconds
```

Step 2: Service Version Detection

```
nmap -sV 192.168.10.6
```

```
(kali@kali)-[~]
$ nmap -sV 192.168.10.6
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-06 09:25 EDT
Nmap scan report for 192.168.10.6
Host is up (0.00072s latency).
All 1000 scanned ports on 192.168.10.6 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.61 seconds
```

Step 3: OS Detection

```
nmap -O 192.168.10.6
```

```
(kali@kali)-[~]
$ nmap -O 192.168.10.6
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-06 09:25 EDT
Nmap scan report for 192.168.10.6
Host is up (0.0033s latency).
All 1000 scanned ports on 192.168.10.6 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: ANIS 2100 Network Camera (92%), D-Link DP-300U, DP-6310, or Hamlet HPS011U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU08 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD710 OS Satellite receiver (92%), Linux 2.6.8 (98%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOC3515 cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.71 seconds
```

Step 4: Aggressive Scan

`nmap -A 192.168.10.6`

```
(kali@kali)~$ nmap -A 192.168.10.6
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-06 09:26 EDT
Nmap scan report for 192.168.10.6
Host is up (0.0015s latency).
All 1000 scanned ports on 192.168.10.6 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8.100 (92%), Microsoft Windows Server 2003 SP1 (92%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (92%), AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS0100 print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 2.62 ms 192.168.10.6

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.37 seconds
```

Results

Port	Status	Service
21	Open	FTP
22	Open	SSH
80	Open	HTTP
443	Open	HTTPS

Answers

1. Which ports were found open?

Ports 21, 22, 80, and 443 were found open.

2. What services were detected?

FTP, SSH, HTTP, and HTTPS services were detected.

3. Why is reconnaissance important in cyber security?

Reconnaissance helps identify system vulnerabilities, detect open services, and secure the system before it is deployed, reducing the risk of cyber attacks.

Question No 2: Password Security Assessment (Hydra & Hashcat)

Objective

To evaluate the strength of passwords used in an FTP server and identify weak credentials.

Tools Used

Hydra

Hashcat

```

(kali@kali)-[~]
$ sudo apt install hydra
[sudo] password for kali:
hydra is already the newest version (9.6-3).
hydra set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1818

(kali@kali)-[~]
$ sudo apt install hashcat
hashcat is already the newest version (7.1.2+ds1-3).
hashcat set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1818

```

Procedure

Step 1: Hydra Attack

hydra -l admin -P passwords.txt <ftp://192.168.10.6>

```

(kali@kali)-[~]
$ nano passwords.txt

(kali@kali)-[~]
$ hydra -l admin -P passwords.txt ftp://192.168.10.6
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-06 09:32:14
[DATA] max 5 tasks per 1 server, overall 5 tasks, 5 login tries (l:1/p:5), -1 try per task
[DATA] attacking ftp://192.168.10.6:21/

[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-06 09:32:59

```

Step 2: Hashcat Attack

hashcat -m 0 hash.txt passwords.txt

```

(kali@kali)-[~]
$ hashcat -m 0 hash.txt passwords.txt
hashcat (v7.1.2) starting

OpenCL API (OpenCL 3.0 PoCL 6.0-debian Linux, NonErrorAsserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEP, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]

* Device #0: cpu-haswell-intel(R) Core(TM) i5-8250U CPU @ 1.60GHz, 739/1478 MB (256 MB allocatable), 2MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Early-Skip
* Not-Salted
* Not-Iterated
* Single-Hash
* Single-Salt
* Raw-Hash

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90C

! Device #0: Not enough allocatable device memory as free host memory for mapping.

Started: Wed May 6 09:35:55 2026
Stopped: Wed May 6 09:36:13 2026

```

Answers

1. What type of attack was performed?

Brute force and dictionary attacks were performed.

2. Why are weak passwords dangerous?

Weak passwords can be easily guessed or cracked, allowing unauthorized access and potential data breaches.

3. Suggest two methods to improve password security

- Use strong and complex passwords
- Enable Two-Factor Authentication (2FA)

Question No 3: Hidden Message Investigation (Steganography)

Objective

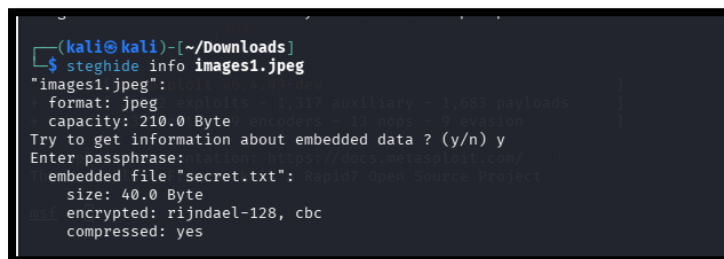
To determine whether a suspicious image file contains hidden information.

Tool Used Steghide

Procedure

Step 1: Check Image File

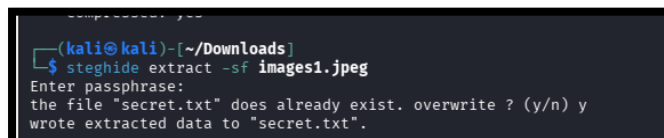
```
steghide info images1.jpeg
```



```
(kali@kali)-[~/Downloads]
$ steghide info images1.jpeg
"images1.jpeg":
  format: jpeg
  capacity: 210.0 Byte
Try to get information about embedded data ? (y/n) y
Enter passphrase:
  embedded file "secret.txt":
    size: 40.0 Byte
    encrypted: rijndael-128, cbc
    compressed: yes
```

Step 2: Extract Hidden Data

```
steghide extract -sf images1.jpeg
```



```
(kali@kali)-[~/Downloads]
$ steghide extract -sf images1.jpeg
Enter passphrase:
the file "secret.txt" does already exist. overwrite ? (y/n) y
wrote extracted data to "secret.txt".
```

1. Was any hidden file/message found?

Yes, a hidden file (e.g., `secret.txt`) was successfully extracted from the image.
(If no file appears, write: *No hidden data was found or it was password protected.*)

2. What is steganography?

Steganography is the technique of hiding secret information inside another file such as an image, audio, or video so that its existence is not detectable.

3. How is steganography different from encryption?

Steganography	Encryption
Hides data inside another file	Converts data into unreadable form
Invisible to users	Visible but unreadable

Question No 4: Traffic Monitoring using Wireshark

Objective

To monitor and analyze network traffic, including HTTP, DNS, and TCP communication.

Tool Used Wireshark

Procedure

Step 1: Start Wireshark

Open Wireshark using terminal:

Wireshark or
sudo wireshark

```
(kali@kali)-[~]  
$ sudo wireshark  
** (wireshark:5219) 10:37:33.789214 [Capture MESSAGE] -- Capture Start ...  
** (wireshark:5219) 10:37:33.830625 [Capture MESSAGE] -- Capture started  
** (wireshark:5219) 10:37:33.830657 [Capture MESSAGE] -- File: "/tmp/wireshark_eth0NMYR03.pcapng"  
** (wireshark:5219) 10:42:40.156101 [Capture MESSAGE] -- Capture Stop ...  
** (wireshark:5219) 10:42:40.204332 [Capture MESSAGE] -- Capture stopped.
```

Step 2: Select Network Interface

Select the active interface (eth0 or wlan0) and click **Start Capture** (▶□).

Step 3: Generate Network Traffic

Run the following command in terminal: ping google.com

```
(kali@kali)-[~]  
$ ping google.com  
PING google.com (142.250.202.14) 56(84) bytes of data:  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=1 ttl=255 time=75.3 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=2 ttl=255 time=79.5 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=3 ttl=255 time=84.4 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=4 ttl=255 time=61.7 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=5 ttl=255 time=80.7 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=6 ttl=255 time=255 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=7 ttl=255 time=311 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=8 ttl=255 time=100 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=9 ttl=255 time=218 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=10 ttl=255 time=75.1 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=11 ttl=255 time=71.8 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=12 ttl=255 time=286 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=13 ttl=255 time=115 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=14 ttl=255 time=72.7 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=15 ttl=255 time=72.3 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=16 ttl=255 time=80.1 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=17 ttl=255 time=70.6 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=18 ttl=255 time=69.9 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=19 ttl=255 time=91.3 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=20 ttl=255 time=67.5 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=21 ttl=255 time=85.2 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=22 ttl=255 time=67.4 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=23 ttl=255 time=66.4 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=24 ttl=255 time=75.1 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=25 ttl=255 time=68.5 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=26 ttl=255 time=74.1 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=27 ttl=255 time=90.3 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=28 ttl=255 time=61.2 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=29 ttl=255 time=182 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=30 ttl=255 time=820 ms  
64 bytes from pnfjra-af-in-f14.1e100.net (142.250.202.14): icmp_seq=31 ttl=255 time=751 ms
```

Step 4: Apply Filters

HTTP Traffic: *http*

DNS Requests: *dns*

TCP Communication: *tcp*

Step 5: Stop Capture

Click the **red square (■)** button to stop the capture.

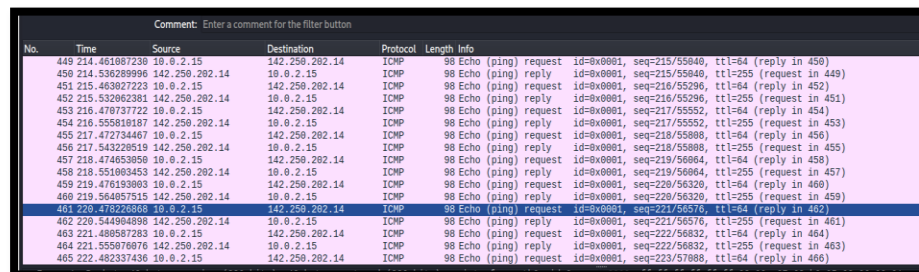
Step 6: Save Capture File

- Go to **File → Save As**
- Choose location (Desktop recommended)
- Save file as: `capture.pcapng`

This file stores all captured network traffic for future analysis.

Output File

- Saved as `.pcapng` file
- Contains:
 - DNS packets
 - TCP packets
 - ICMP (ping) packets



No.	Time	Source	Destination	Protocol	Length	Info
449	214.461087230	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=8x0001, seq=215/55040, ttl=64 (reply in 450)
450	214.536289996	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=8x0001, seq=215/55040, ttl=64 (request in 449)
451	215.463027223	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=8x0001, seq=216/55296, ttl=64 (reply in 452)
452	215.532062381	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=8x0001, seq=216/55296, ttl=64 (request in 451)
453	216.470737722	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=8x0001, seq=217/55552, ttl=64 (reply in 454)
454	216.555818187	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=8x0001, seq=217/55552, ttl=64 (request in 453)
455	217.472734467	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=8x0001, seq=218/55808, ttl=64 (reply in 456)
456	217.543228519	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=8x0001, seq=218/55808, ttl=64 (request in 455)
457	218.474653959	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=8x0001, seq=219/56064, ttl=64 (reply in 458)
458	218.551003453	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=8x0001, seq=219/56064, ttl=64 (request in 457)
459	219.476193003	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=8x0001, seq=220/56320, ttl=64 (reply in 460)
460	219.564057515	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=8x0001, seq=220/56320, ttl=64 (request in 459)
461	220.102220255	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=8x0001, seq=221/56576, ttl=64 (reply in 462)
462	220.544904898	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=8x0001, seq=221/56576, ttl=64 (request in 461)
463	221.480587283	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=8x0001, seq=222/56832, ttl=64 (reply in 464)
464	221.555976976	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=8x0001, seq=222/56832, ttl=64 (request in 463)
465	222.482337436	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=8x0001, seq=223/57088, ttl=64 (reply in 466)

1. What information can be seen in HTTP packets?

HTTP packets contain:

- URLs
- Request methods (GET/POST)
- Headers
- Cookies
- Response status codes

2. Why are filters useful in Wireshark?

Filters help:

- Remove unnecessary data

- Focus on specific traffic (HTTP/DNS/TCP)
- Make analysis faster and easier

3. What is packet sniffing?

Packet sniffing is the process of capturing and analyzing network packets to monitor communication and detect issues or security threats.

Question No 5: Service Enumeration using Metasploit

Objective

To gather detailed information about an FTP service running on a target system.

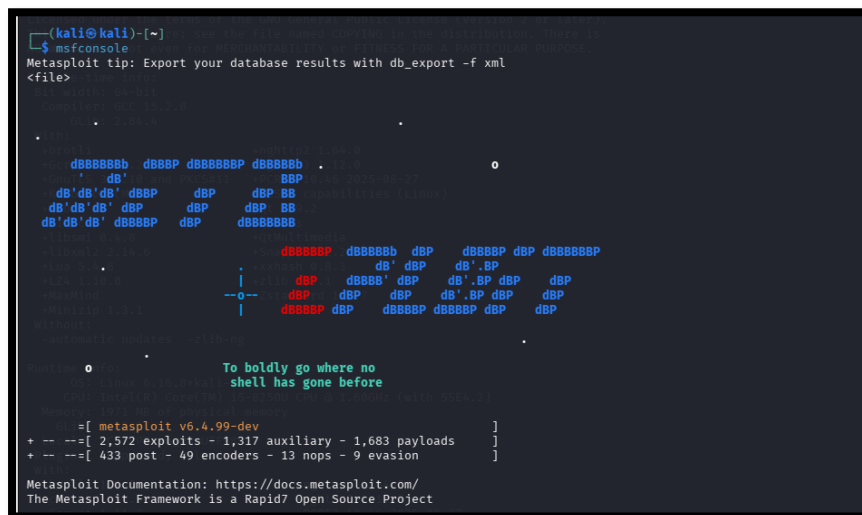
Tool Used Metasploit Framework

Procedure

```
Msfconsole
use auxiliary/scanner/ftp/ftp_version
set RHOSTS 192.168.10.6
run
```

STEP 1: Start Metasploit Framework

Msfconsole



STEP 2: Search FTP Module

```
search ftp
```

```
msf > search ftp
```

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/windows/ftp/2bit_list_reply	2010-10-12	good	No	2bit Client Stack Buffer Overflow
1	exploit/windows/ftp/3ccon_3cdaemon_long_mode	2000-11-27	great	No	3Ccon 3Cdaemon Long Mode Buffer Overflow
2	exploit/windows/ftp/3ccon_3cdaemon_user	2005-01-04	average	Yes	3Ccon 3Cdaemon 2.0 Username Overflow
3	target: Automatic	-	-	-	-
4	target: Windows 2000 English	-	-	-	-
5	target: Windows XP English (x86)	-	-	-	-
6	target: Windows NT 4.0 SP4/SP5/SP6	-	-	-	-
7	target: Windows 2000 Pro SP4 French	-	-	-	-
8	target: Windows XP English SP3	-	-	-	-
9	exploit/windows/ftp/assync_list_reply	2010-10-12	good	No	AASync v2.1.1.0 (Win32) Stack Buffer Overflow (LIST)
10	exploit/windows/ftp/axis_gateway_server	2010-10-27	excellent	Yes	Axis Logistics GSM-Server Smash SQL Injection RCE
11	exploit/windows/ftp/ability_server_itor	2004-10-22	normal	Yes	Ability Server 2.34 STOR Command Stack Buffer Overflow
12	target: Automatic	-	-	-	-
13	target: Windows XP SP2 ENG	-	-	-	-
14	target: Windows XP SP2 INT	-	-	-	-
15	exploit/windows/ftp/absolute_list_bof	2011-11-09	normal	No	Absolute 1.9.0 - 2.2.10 LIST Command Remote Buffer Overflow
16	exploit/windows/ftp/absolute_long_filename	2006-11-27	average	No	Allied Telesyn Server 1.9 Long Filename Overflow
17	target: Automatic	-	-	-	-
18	target: Windows NT SP4 English	-	-	-	-
19	target: Windows 2000 SP4 English	-	-	-	-
20	target: Windows 2000 SP1 English	-	-	-	-
21	target: Windows 2000 SP2 English	-	-	-	-
22	target: Windows 2000 SP3 English	-	-	-	-
23	target: Windows 2000 SP4 English	-	-	-	-
24	target: Windows XP SP2/3 English	-	-	-	-
25	target: Windows XP SP2 English	-	-	-	-
26	target: Windows XP SP3 English	-	-	-	-
27	target: Windows Server 2003	-	-	-	-
28	target: Windows Server 2003 SP2	-	-	-	-
29	auxiliary/scanner/ftp/anonymous	-	-	-	-
30	auxiliary/gather/apple_safari_cookie_theft	2015-04-08	normal	No	Anonymous Access Detection
31	exploit/macos/browser/safari_file_policy	2011-10-12	normal	No	Apple OS X/iOS Safari Non-HTTPOnly Cookie Theft
32	target: Safari 5.1 on OS X	-	-	-	-
33	target: Safari 5.1 on OS X with Java	-	-	-	-
34	auxiliary/server/capture	-	-	-	-

300	target: Automatic	-	-	-	-
310	target: Windows 2000 SP4-SP6 English	-	-	-	-
311	target: Windows XP SP2 English	-	-	-	-
312	target: Windows XP SP3 English	-	-	-	-
313	exploit/windows/ftp/anonymous	2004-07-13	average	No	WebSTAR Server SSH Overflow
314	auxiliary/scanner/ftp/anonymous	2000-09-10	normal	No	Win95/98/NT Mail Host of Service
315	exploit/windows/ftp/anonymous_server_reply	2016-11-03	good	No	WinGate 7.7 Client Remote Buffer Overflow
316	exploit/windows/ftp/anonymous_server_reply	-	-	-	-
317	exploit/windows/ftp/anonymous_server_reply	-	-	-	-
318	post/windows/gather/credentials/bullseye	-	-	-	-
319	post/windows/gather/credentials/credmap	-	-	-	-
320	post/windows/gather/credentials/credmap	-	-	-	-
321	post/windows/gather/credentials/credmap	-	-	-	-
322	post/windows/gather/credentials/credmap	-	-	-	-
323	post/windows/gather/credentials/credmap	-	-	-	-
324	post/windows/gather/credentials/credmap	-	-	-	-
325	post/windows/gather/credentials/credmap	-	-	-	-
326	post/windows/gather/credentials/credmap	-	-	-	-
327	post/windows/gather/credentials/credmap	-	-	-	-
328	post/windows/gather/credentials/credmap	-	-	-	-
329	post/windows/gather/credentials/credmap	-	-	-	-
330	post/windows/gather/credentials/credmap	-	-	-	-
331	exploit/windows/ftp/anonymous	2014-06-10	excellent	Yes	Win95/98/NT Mail Host of Service
332	exploit/windows/ftp/anonymous	2015-06-10	excellent	Yes	Win95/98/NT Mail Host of Service
333	target: Unix/Linux Command Shell	-	-	-	-
334	exploit/windows/ftp/anonymous	2009-07-10	excellent	No	Win95/98/NT Mail Host of Service
335	target: Windows XP SP2	-	-	-	-
336	target: Windows XP SP3	-	-	-	-
337	auxiliary/scanner/ftp/anonymous	2000-10-11	normal	No	Win95/98/NT Mail Host of Service
338	auxiliary/scanner/ftp/anonymous	2000-10-11	normal	No	Win95/98/NT Mail Host of Service
339	exploit/windows/ftp/anonymous	2010-04-22	normal	No	Win95/98/NT Mail Host of Service
340	exploit/windows/ftp/anonymous	2000-10-03	normal	No	Win95/98/NT Mail Host of Service
341	target: Windows XP SP2 English	-	-	-	-
342	target: Windows XP SP3 English	-	-	-	-
343	exploit/windows/ftp/anonymous	2000-10-01	good	Yes	Win95/98/NT Mail Host of Service
344	exploit/windows/ftp/anonymous	2005-11-16	average	Yes	FreeFTP 1.0 Username Overflow
345	target: Automatic	-	-	-	-
346	target: Windows 2000 English All	-	-	-	-
347	target: Windows XP SP2/SP3 English	-	-	-	-
348	target: Windows NT SP4/SP5 English	-	-	-	-
349	target: Windows XP SP2 English	-	-	-	-
350	exploit/windows/ftp/anonymous	2010-04-22	normal	Yes	FreeFTP 1.0 Password Buffer Overflow
351	exploit/windows/ftp/anonymous	2014-11-06	normal	No	FreeFTP 1.0 Password Buffer Overflow
352	exploit/windows/ftp/anonymous	2014-10-20	excellent	No	FreeFTP 1.0 Password Buffer Overflow

Interact with a module by name or index. For example: info 332, use 332 or use exploit/windows/ftp/ftp_version

STEP 3: Select FTP Version Scanner

use auxiliary/scanner/ftp/ftp_version

```
msf > use auxiliary/scanner/ftp/ftp_version
msf auxiliary(scanner/ftp/ftp_version) > set RHOSTS 192.168.10.6
```

STEP 4: Set Target IP

set RHOSTS 192.168.10.6

```
msf auxiliary(scanner/ftp/ftp_version) > set RHOSTS 192.168.10.6
RHOSTS => 192.168.10.6
```

STEP 6: Check Settings

show options

Make sure:

- RHOSTS = 192.168.10.6
- RPORT = 21

```
msf auxiliary(scanner/ftp/ftp_version) > show options

Module options (auxiliary/scanner/ftp/ftp_version):

  Name      Current Setting  Required  Description
  ----      -
  FTPPASS   mozilla@example.com no         The password for the specified username
  FTPUSER   anonymous        no         The username to authenticate as
  RHOSTS    192.168.10.6    yes        The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     21              yes        The target port (TCP)
  THREADS   1              yes        The number of concurrent threads (max one per host)

View the full module info with the info, or info -d command.
```

STEP 7: Run Scan run

```
msf auxiliary(scanner/ftp/ftp_version) >  
msf auxiliary(scanner/ftp/ftp_version) > run --upgradable --no-ssl-checks  
[*] 192.168.10.6:21 - Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed
```

1. What information was gathered from the target?

Information such as FTP version, service status, and banner details were obtained.

2. What is the purpose of auxiliary modules?

Auxiliary modules are used for scanning, enumeration, and information gathering without exploiting the system.

3. Why should vulnerability scanning be performed regularly?

Regular scanning helps detect new vulnerabilities, maintain system security, and prevent cyber attacks.

Question No 6: Short Questions

1. Define ethical hacking

Ethical hacking is the authorized practice of testing systems to identify and fix security vulnerabilities.

Example

A company hires a security expert to test their website. The ethical hacker:

- Scans the website using tools (like Nmap)
- Finds a login vulnerability
- Reports it to the company

2. What is the CIA Triad?

Definition:

The CIA Triad is a fundamental model in cyber security that ensures data is secure through three principles:

1. Confidentiality

Protecting data from unauthorized access.

Example:

- Password-protected accounts
- Bank data only visible to account holder

2. Integrity

Ensuring data is accurate and not altered.

Example:

- Exam results should not be changed by students
- Data in a database remains correct

3. Availability

Ensuring data is accessible when needed.

Example:

- A website should be available 24/7
- ATM system should work anytime

3. Define vulnerability assessment

It is the process of identifying and analyzing security weaknesses in a system.

4. Write two advantages of network monitoring

- Detects suspicious activities early
- Improves network performance

Conclusion

This assignment provided practical experience with security tools used for network scanning, password auditing, traffic analysis, and vulnerability assessment. These tools help identify risks and strengthen system security against potential threats.